

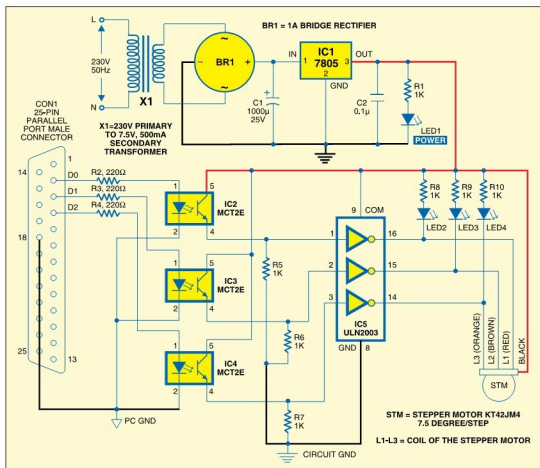


Fig. 3: Circuit of stepper motor controller using 'C'

PARTS LIST

Semiconductors:

- IC1 - 7805 5V regulator
- IC2-IC4 - MCT2E optocoupler
- IC5 - ULN2003 buffer/driver
- BR1 - 1A bridge rectifier

Resistors (all 1/4-watt, ±5% carbon):

- LED1-LED4 - 5mm light-emitting diode
- R1, R5-R10 - 1-kilo-ohm
- R2-R4 - 220-ohm

Capacitors:

- C1 - 1000µF, 25V electrolytic
- C2 - 0.1µF ceramic

Miscellaneous:

- CON1 - 25-pin, D-type, parallel-port male connector
- X1 - 230V AC primary to 7.5V, 500mA secondary transformer
- STM - KT42JM4, 7.5 degrees/step, 5V stepper motor

In this project, only data port pins 2 through 4 have been used to control the stepper motor. Fig. 2 shows author's prototype of the

the stepper motor to the parallel port. Each of these ICs has an internal LED and a photo-transistor. The LED glows and the photo-transistor conducts as explained in the succeeding section. However, the external circuitry of the LED and photo-transistor differ. The LED is driven by the signal from the parallel port, whereas a separate 5V supply powers the photo-transistor. Therefore it completely isolates the parallel port from rest of the interfacing circuit. This ensures safety of the parallel port against damage from shorting of the external circuit.

IC ULN2003 (IC5) is used to

drive the stepper motor. It is a high-voltage, high-current Darlington transistor array. It consists of seven npn Darlington pairs that feature high-voltage outputs with internal common-cathode clamp diodes for switching inductive loads. The collector-current rating of a Darlington pair is 500 mA. The Darlington pairs can be paralleled for higher current capability. This IC can also be used as a relay driver, lamp driver, display driver, line driver or logic buffer.

When a high pulse ('1') is applied to pin 1 of IC2 (MCT2E), its internal LED glows and the corresponding photo-transistor starts conducting through the external circuit. The output of IC2 at pin 4 is fed to pin 1 of IC5. Pin 16 is connected to L1 coil of the stepper motor. Similarly, the output pins of IC3 and IC4 are connected to pins 2 and

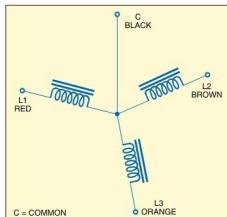


Fig. 4: Lead identification of KT42JM4 stepper motor

stepper motor controller.

Circuit description

Fig. 3 shows a simple interfacing circuit to control the stepper motor. Optocoupler ICs MCT2E (IC2 through IC4) are recommended for interfacing